Doruk Karademirler - RESUME

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portfolio-ashy-two-53.vercel.app

EDUCATION

University of Toronto | Honors Bachelor of Science **Sep. 2021 - Nov. 2026 (Expected)**
Computer Science Specialist, Economics Minor, 3.50 GPA Mississauga/Toronto, ON

- Participated in the PEY Co-op Program.
- Relevant courses: Computer Security | Introduction to Information Security | Principles of Computer Networks | Software Testing and Verification | Computational Complexity and Computability | Algorithm Design and Analysis | Software Tools and Systems Programming | Data Structures and Analysis | Introduction to Software Engineering | Introduction to Databases.

PROFESSIONAL STATEMENT / SUMMARY

Computer science student with 6 years of experience in computer-aided design and developing problem-solving algorithms. Proficient in programming languages such as Java, JavaScript, Python, SQL, CSS, and C. Skilled in utilizing Git for version control and developing CI/CD pipelines for compiler toolchains. Experienced in using technical expertise to analyze and interpret economic data.

WORK EXPERIENCE

SOFTWARE TOOLCHAINS ENGINEERING INTERN **July 2024- September 2025**
Qualcomm Markham, ON

- Diagnosed and resolved a performance bottleneck in the expf.c function of Android's math library, eliminating 4K page crossing and improving HDR value consistency. Achieved a 28% performance boost in Geekbench benchmarks.
- Designed and implemented a comprehensive **CI/CD pipeline** using GitLab to automate build, test, and publish stages for GCC and LLVM toolchains. Created and configured **virtual machines** to validate pipeline functionality and benchmark performance using **SPEC CPU 2017**, ensuring consistent and scalable compiler optimization across AArch64 and X86 targets.
- Implemented a pointer chase optimization algorithm for both GCC and LLVM compilers, reducing execution cycles for load instructions that reuse the same register and improving memory access efficiency. This resulted in a 3% performance gain in SPEC benchmarks with pointer-heavy data structures.

TEACHING ASSISTANT **Sept. 2023 - January 2024**
UNIVERSITY OF TORONTO Mississauga, ON

- Conducted 6 individual tutorial sections (40 students per session) and office hours for CSC236 - Introduction to the Theory of Computation, achieving a 15% improvement in student performance on course assessments.
- Responsibly graded assignments and exams, ensuring timely feedback to students while maintaining consistent grading standards.
- Collaborated closely with the course instructor to align teaching strategies with the course objectives, resulting in a cohesive learning experience.

PROJECTS

Price of Empire (Research Opportunity Program) | Visual Studio Code, R, Python **Summer 2023**
The research project focuses on creating a comprehensive meta-database accessible through the UTM Economics website, providing researchers with detailed Canadian trade data. Under the guidance of Professor Nicholas Zammit, developed a robust Python algorithm to cleanse and structure a complex economic dataset using the pandas library, which will be used to analyze the British colonial relationships on Dominion countries like Canada. Worked closely with a diverse team of economists and computer scientists to merge domain-specific knowledge with technical approach.

Software Developer - CourseBind Website | ReactJS, CSS, PHP, JavaScript, SQL **January 2024**
As a key member of a five-person development team, contributing to the design, development, and deployment of CourseBind, a comprehensive educational platform designed to enhance student productivity and learning. Demonstrated proficiency in full-stack web development, from conceptualization to deployment, within a collaborative, Agile team environment.

Security Vulnerability Analysis & Remediation (CSC347) | C, OpenSSL, Bash, Perl **October 2025**

Analyzed a vulnerable C account-management application, identifying and documenting 15 security vulnerabilities spanning path traversal, buffer handling, authentication, password storage, and transaction validation. Developed and tested fixes for 11 issues using input validation, safer string handling, account identity verification, password hashing, and file-operation error handling, then reproduced each vulnerability in a controlled environment to assess impact against the CIA triad.

CERTIFICATIONS

- Google Cybersecurity Professional Certificate, Google | Issued Oct. 2026 | Credential ID: MYTC3W2E8NGA